

Chemistry Ch 4.5: Transition Elements

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Periodic table ke **darmiyan** wala chaura hissa. Ye **d block** ke elements hain: iron, copper, zinc, nickel, chromium.

Paanch khasoosiyat, ye hi poochhi jati hain

1. **Oonchi density.**
2. **Oonche melting points.**
3. **Mukhtalif oxidation states** (variable), yani ek hi dhat alag alag charge le sakti hai.
4. **Rangeen compounds** banate hain.
5. **Catalysts** ke tor par kaam karte hain.

Ghar se misalein

Variable oxidation states: iron Fe^{2+} bhi banata hai aur Fe^{3+} bhi. Isi liye zang ka rang alag alag hota hai.

Rangeen compounds: copper ke compounds **neele sabz**, aur isi liye purani tambe ki chhat sabz par jati hai.

Catalysts: Haber's process mein **iron**, aur contact process mein vanadium.

Rang aur catalyst, dono ki ek hi wajah

Ye dono **d sub-shell** ke adhoore bharne se aate hain. Adhoori d sub-shell electrons ko aane jaane deti hai, isi se rang bhi banta hai aur catalyst wala kaam bhi hota hai.

Quiz

1. Name the **five** characteristic properties of the transition elements your book gives.
2. Give one everyday example of a transition metal showing variable oxidation states, and one of a transition metal acting as a catalyst.